Device-Level Characterization of Back-Gate-Tunable Zero-Change MOSFETs for AI-Oriented Silicon Photonics

Yunfeng Li[a,†], Dezheng Xu[a,†], Zhongyi Pan[a,†], Shuo Lin[a], Yuting Zhang[a], Weijie Hong[b], and Hongtao Lin[a,b,*]
[a]*The State Key Lab of Brain-Machine Intelligence, Key Laboratory of Micro-Nano Electronics and Smart System of Zhejiang Province, College of Information Science and Electronic Engineering, Zhejiang University*, Hangzhou 310027, China
[b]*College of Integrated Circuits, Zhejiang University*, Hangzhou 310027, China
{3230101966, 3230105646, 3230105795, 12531153, 22431143, 22547118, hometown}@zju.edu.cn
†Yunfeng Li, Dezheng Xu, and Zhongyi Pan share first authorship and are listed in order of contribution.
*Corresponding author: Hongtao Lin (hometown@zju.edu.cn).

***Abstract*—Artificial intelligence (AI)-oriented silicon-photonic systems require distributed electronic interfaces for weight and phase setting, monitor selection, and repeated calibration. Zero-change electronic devices can place these functions on chip without adding transistor-specific process modules; however, MOSFETs implemented in photonic foundry processes may exhibit zero-back-gate operating points outside the gate-voltage window preferred for complementary control circuits. We characterize complementary side-gate MOSFETs fabricated without additional process modules on a 220-nm silicon-on-insulator (SOI) platform and quantify their threshold response to a global back-gate bias. Using a common 100-nA fixed-current definition, the nMOS threshold shifts from −10.20 to +3.10 V, corresponding to a 13.30-V span that crosses zero. The pMOS threshold shifts monotonically from −4.59 to −1.64 V over the measured substrate-bias range. The measured tuning establishes the shared substrate as an effective die-level control knob for placing zero-change MOSFETs within operating windows suitable for low-speed, high-impedance monitor selection, phase setting, and calibration interfaces in AI-oriented photonics.**

***Index Terms*—silicon photonics, side-gate MOSFET, complementary MOS, back-gate bias, threshold tuning, AI photonics**

I. Introduction

Artificial intelligence (AI)-oriented silicon-photonic processors based on Mach–Zehnder-interferometer meshes and programmable optical networks require distributed electronic interfaces for weight or phase programming, monitor selection, and repeated calibration [1]–[4]. Their replication across many channels creates substantial demands on control density and off-chip I/O. Zero-change electronics can embed these functions in silicon-photonic chips without altering the foundry process flow [5], [6]. However, the 220-nm silicon-on-insulator (SOI) process is optimized primarily for photonic components rather than transistor electrostatics. Restricted doping options, the absence of a self-aligned gate process, and the lateral side-gate geometry leave transistor thresholds largely determined by the native device-layer doping, interface charge, and process-defined dimensions. Consequently, the zero-substrate-bias operating points of complementary devices are not inherently aligned with the gate-voltage window required by electronic control circuits.

This operating-point mismatch is also evident in previous zero-change circuit demonstrations. Zanetto *et al.* implemented complementary side-gate MOSFETs, a 16-to-1 multiplexer, time-multiplexed photodiode readout, sample-and-hold control of thermal actuators, and programmable-interferometer configuration in a conventional 220-nm silicon-photonics flow [5], [7]. In that work, the chip substrate served as a common back gate, and a substrate voltage of $V_{\mathrm{SUB}} = -10$ V was selected to obtain favorable operation and low leakage for both transistor polarities [5]. These demonstrations established the circuit-level utility of the devices at a chosen bias but did not provide a systematic map of the complementary threshold trajectories, accessible tuning ranges, or polarity-dependent sensitivities.

Back-gate control provides a natural post-fabrication degree of freedom for addressing this gap. In an SOI stack, the silicon substrate couples electrostatically to the thin device layer through the buried oxide, allowing the channel potential and threshold voltage to be adjusted after fabrication [8], [9]. The central device-level questions are therefore quantitative: over what range can the nMOS and pMOS thresholds be shifted, and how differently do the two polarities respond to a common substrate voltage? Quantifying these responses turns the substrate bias from an empirical circuit setting into a measurable device-design parameter. Properly biased zero-change MOSFETs can then support high-impedance monitor multiplexers, sample-and-hold nodes, bias selection, and slow feedback in close proximity to the photonic devices.

Here, we present room-temperature characterization of the zero-back-gate operation and global back-gate response of complementary side-gate MOSFETs using a common 100-nA fixed-current threshold definition. At $V_{\mathrm{BG}} = 0$ V, the nMOS and pMOS thresholds are −10.201 and −4.594 V, respectively. Applying a negative substrate bias shifts the nMOS threshold through zero over a 13.30-V span and moves the pMOS threshold monotonically over 2.96 V. These measurements provide a complementary back-gate tuning map and clarify how the shared substrate can be used to position zero-change devices for low-speed AI-photonic configuration and calibration interfaces.

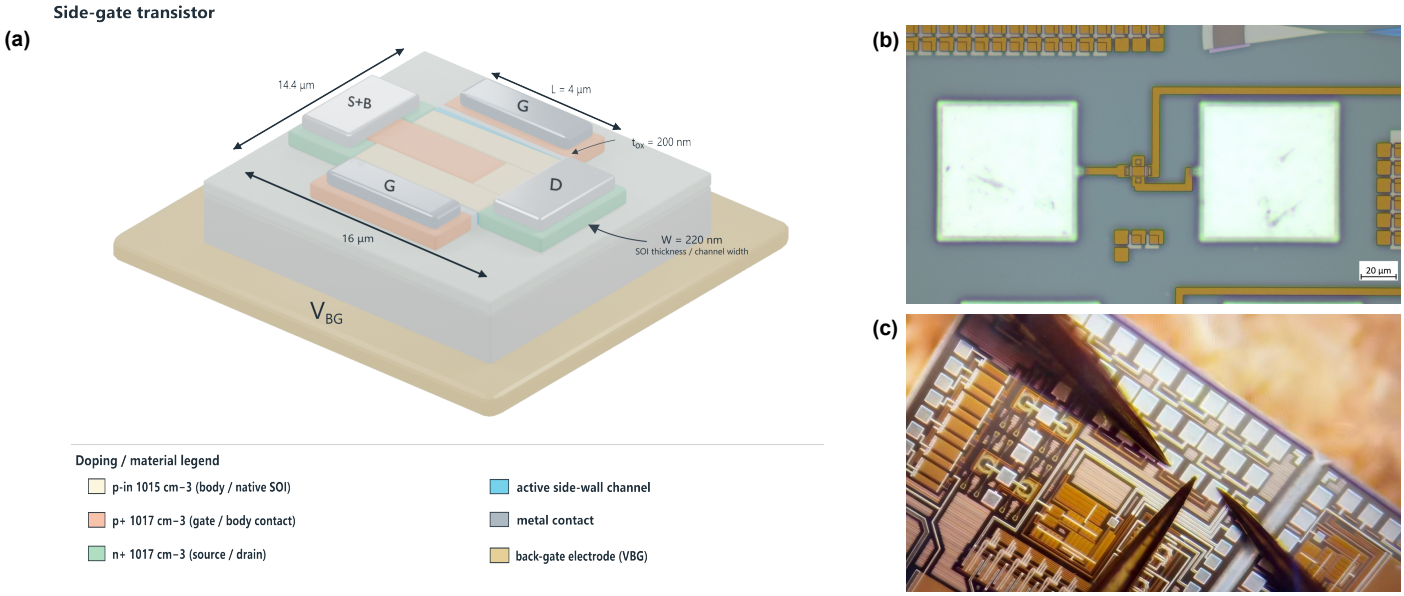


Fig. 1. Side-gate device structure and measurement views. (a) Author-drawn schematic of the side-gate transistor and common back-gate geometry; the vertical dimension is exaggerated for clarity. (b) Optical micrograph of a measured device structure. (c) Probe contact during electrical characterization. In (a), B denotes the local body terminal, S+B denotes the combined source/body contact, and V_{BG} denotes the global back-gate bias.

II. Device Structure and Characterization

A. Zero-Change Side-Gate Devices

In this work, zero-change refers to devices implemented entirely with the masks, implants, and anneal modules already available in the silicon-photonic platform, without any transistor-specific process additions. The devices adopt the side-gate geometry developed for a standard silicon-photonics design flow [5], as illustrated in Fig. 1(a). The nominal channel length is $L = 4\ \mu$m, the lateral gate-to-channel spacing is 200 nm, and the 220-nm silicon device-layer thickness defines the channel width. A buried-oxide layer of thickness 3 μm separates the silicon device layer from the substrate. The local body terminal B is connected to the source as S+B. This local body contact is electrically distinct from the common silicon substrate used as the global back gate. Complementary nMOS and pMOS devices are obtained by interchanging the relevant source/drain and body-contact doping polarities. The source-connected local body fixes the device-level body potential, whereas the substrate couples through the buried oxide to provide common-mode electrostatic control across the die. The optical micrograph in Fig. 1(b) shows the measured device structure.

B. Measurement Configuration

Electrical measurements were performed in ambient air at room temperature using an Fs-Pro semiconductor parameter analyzer connected to the probe station shown in Fig. 1(c). The drain current is reported as $|I_D|$ to provide a common sign convention for the complementary devices.

The back-gate voltage is defined as

$$V_{\mathrm{BG}} = V_{\mathrm{SUB}} - V_S, \tag{1}$$

where V_{SUB} is the potential of the common silicon substrate. Because the source was grounded throughout the measurements reported here, the instrument label V_b is numerically identical to V_{BG}. Under this convention, the lateral gate provides local device excitation, whereas the silicon substrate provides shared back-gate control.

III. Results and Discussion

A. Zero-Bias Baseline and Complementary Device Operation

The zero-back-gate baseline was established in a conventional three-terminal configuration with both the source and substrate grounded. For the transfer measurements, the nMOS gate was swept from −14 to −8 V at drain voltages from 1 to 5 V, whereas the pMOS gate was swept from 0 to −6 V at drain voltages from −1 to −5 V. The output characteristics were recorded using drain sweeps from 0 to 5 V for nMOS and from −5 to 0 V for pMOS.

The transfer characteristics in Fig. 2(a) and Fig. 2(b) exhibit systematic gate modulation and well-defined transitions between the low- and higher-current regimes for both polarities. The output characteristics in Fig. 2(c) and Fig. 2(d) are approximately linear in the low-drain-voltage regime. Together, these results confirm complementary field-effect operation at $V_{\mathrm{BG}} = 0$ V. At the same time, the markedly negative gate-voltage ranges required for conduction reveal that the process-native operating points are substantially displaced from zero.

Throughout this work, the threshold voltage is defined by the $|I_D| = 100$ nA fixed-current criterion. For each transfer curve, V_{th} is obtained by linearly interpolating V_G between the two adjacent measured points that bracket 100 nA on the $\log_{10}|I_D|$ axis. The target current lies within every analyzed

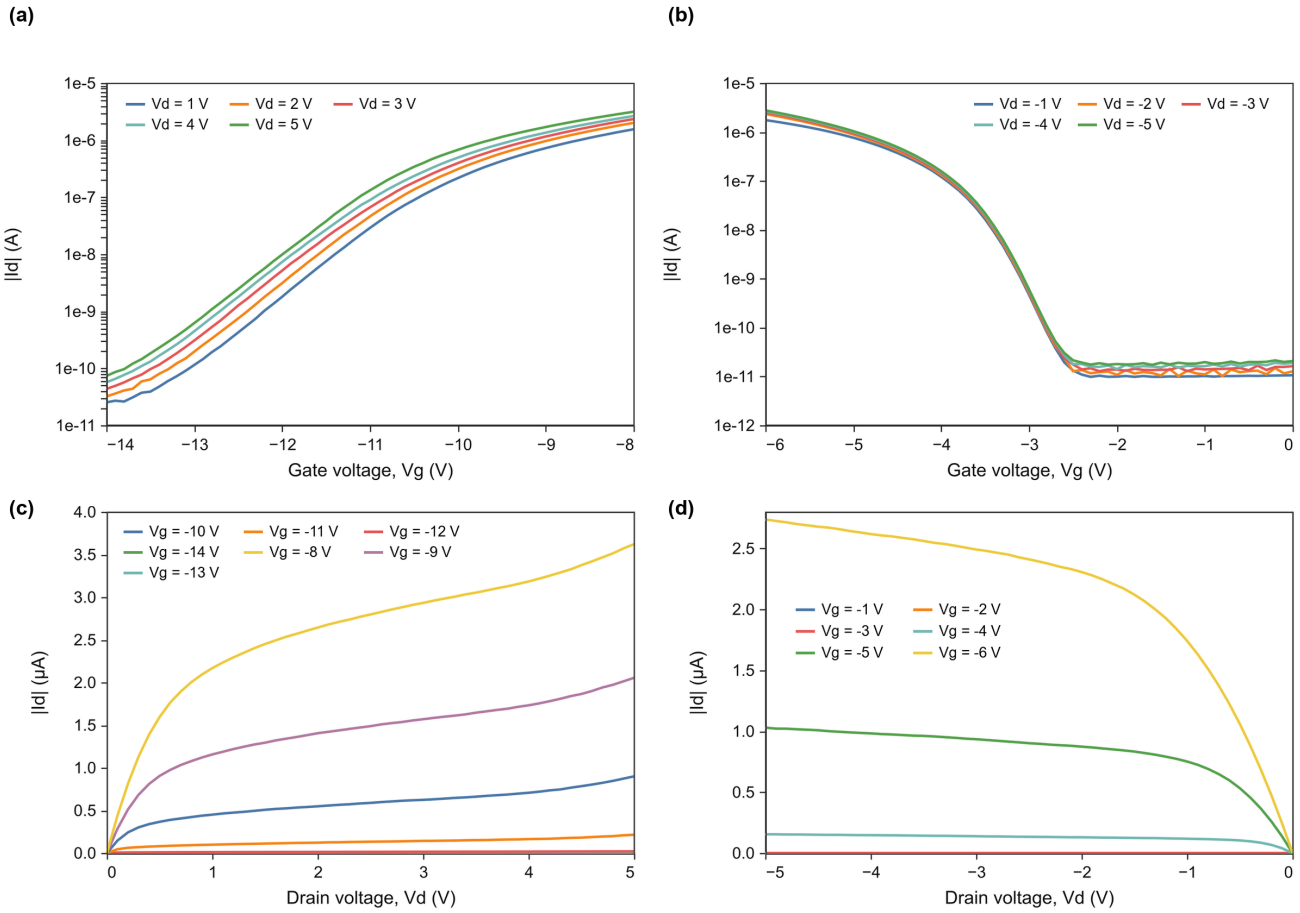


Fig. 2. Measured complementary MOSFET characteristics. (a) nMOS transfer curves for V_D = 1–5 V. (b) pMOS transfer curves for V_D = −1 to −5 V. (c) nMOS output curves for V_G = −14 to −8 V. (d) pMOS output curves for V_G = −1 to −6 V.

TABLE I
EXTRACTED PARAMETERS OF THE MEASURED DEVICES.

Parameter	nMOS	pMOS
V_{th} (V)	−10.20	−4.59
SS (mV/dec)	851.7	302.5
DIBL (mV/V)	170.7	20.4
I_{on} (μA)	3.153	2.770
I_{off} (A)	7.375×10^{-11}	2.072×10^{-11}
I_{on}/I_{off}	4.275×10^{4}	1.336×10^{5}
R_{on} (kΩ)	313.9	472.0

transfer curve and therefore enables consistent extraction without numerical differentiation. SS was calculated as the inverse magnitude of the slope from a linear regression of $\log_{10}|I_D|$ versus V_G over a continuous nine-point subthreshold interval spanning at least 0.5 current decade. DIBL was calculated from the difference between the 100-nA thresholds measured at $|V_D|$ = 1 and 5 V. The on- and off-state currents were evaluated at the measured gate-voltage endpoints, and R_{on} was extracted from a linear fit over the first 10% of the output-voltage range.

Table I summarizes the extracted parameters of the measured devices. At zero back-gate bias, both thresholds lie outside a compact complementary gate-voltage window, with the nMOS exhibiting the larger negative displacement. Nevertheless, the measured microampere-scale on-currents and on/off ratios are compatible with switching and selection at high-impedance nodes, including monitor multiplexing and bias routing; these functions are distinct from high-current drive. For the intended control applications, the position of the operating window is therefore a primary device-design consideration.

B. Global Back-Gate Threshold Modulation

Four-terminal back-gate transfer characteristics were recorded at $|V_D|$ = 0.1 V using substrate biases from −70 to −30 V for nMOS and from −35 to 0 V for pMOS; the nMOS zero-bias point was obtained from the conventional three-terminal baseline. These voltages were applied to the global silicon substrate rather than to the local lateral gate. Because the substrate couples to the device layer through the 3 μm-thick buried oxide, a wider substrate-bias range was required to capture the nMOS threshold crossing and map the full tuning response. The applied values should therefore be interpreted as device-characterization biases for the substrate terminal rather than as the local gate or circuit-supply range.

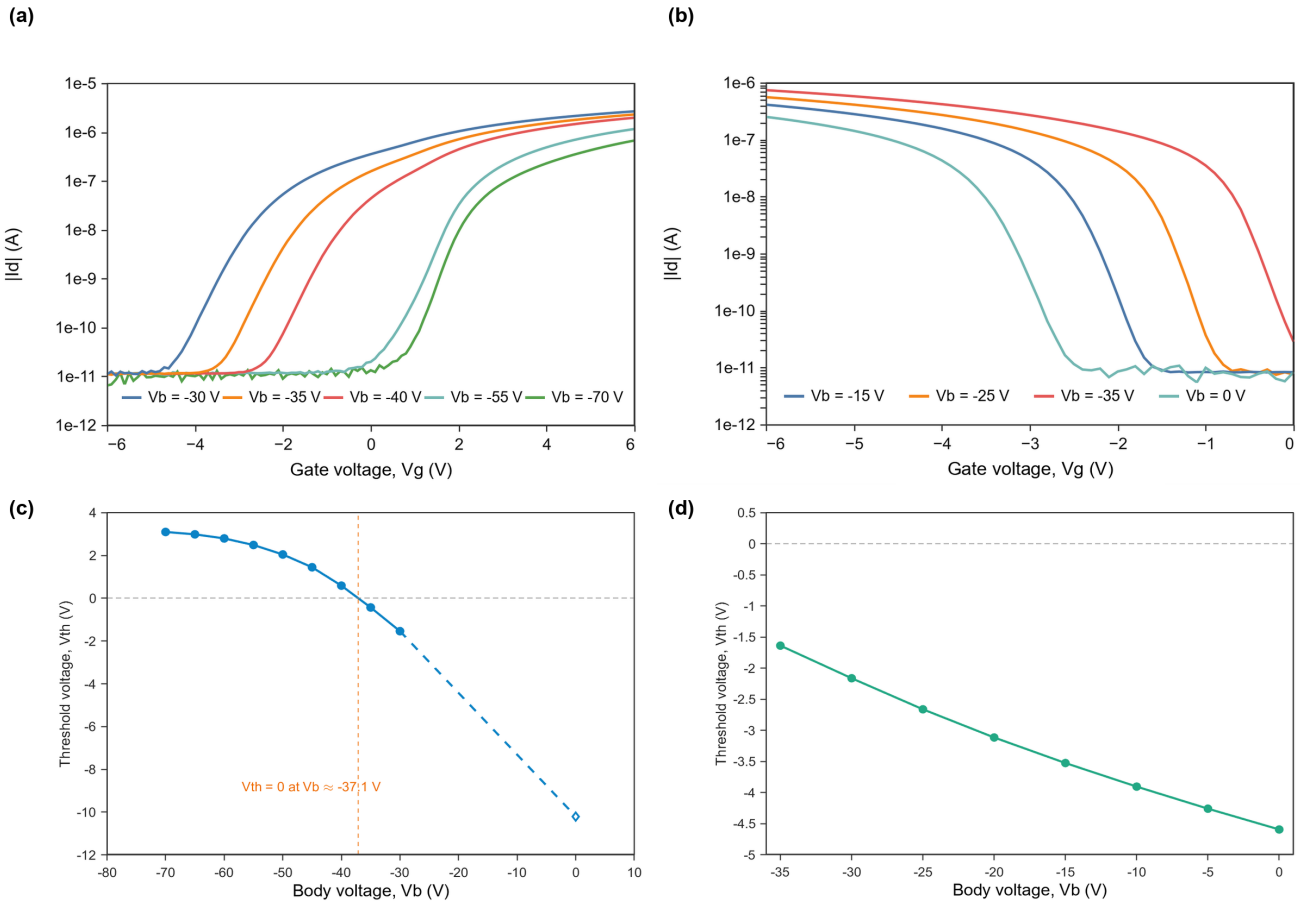


Fig. 3. Back-gate modulation of the measured devices at $|V_D| = 0.1$ V. (a,b) Measured nMOS and pMOS transfer characteristics at selected substrate biases. (c,d) Threshold voltages obtained using the common $|I_D| = 100$ nA fixed-current criterion. With the source grounded, the instrument label V_b is equal to V_{BG}. The nMOS zero-bias point is taken from the conventional three-terminal scan and is connected to the four-terminal −30 V point by a dashed line because no intermediate substrate biases were measured. The line joining the pMOS points is a guide to the measured trend.

The transfer curves in Fig. 3(a) and the corresponding 100-nA thresholds in Fig. 3(c) show that the nMOS threshold shifts from −10.201 V at $V_{BG} = 0$ V to +3.098 V at $V_{BG} = -70$ V, yielding a measured span of 13.299 V. The threshold changes sign between the measured −40 and −35 V points, and interpolation between these neighboring values places the crossing near $V_{BG} = -37.1$ V. The shared substrate thus provides continuous post-fabrication adjustment from a strongly negative threshold through zero and into positive-threshold operation.

For pMOS, the transfer characteristics in Fig. 3(b) and the threshold points in Fig. 3(d) show a monotonic shift from −4.594 V at $V_{BG} = 0$ V to −1.637 V at $V_{BG} = -35$ V, corresponding to a 2.956-V span. A negative substrate bias therefore moves both transistor polarities toward a more compact complementary gate-voltage window. The different nMOS and pMOS tuning magnitudes, however, require both polarities to be considered when selecting a common substrate bias.

These polarity-dependent threshold shifts can be understood qualitatively as electrostatic coupling from the shared substrate, through the buried oxide, to the thin silicon device layer. Changing V_{SUB} therefore changes the channel potential and the lateral-gate voltage required to reach the 100-nA conduction condition. In the present side-gate geometry, the different nMOS and pMOS tuning magnitudes may reflect the combined effects of gate/back-gate capacitive coupling, polarity-dependent doping, interface charge, and contact geometry.

C. Implications for AI-Oriented Photonic Control

Photonic neural networks and matrix processors based on Mach–Zehnder-interferometer meshes require electronic functions for weight or phase programming, monitor selection, and repeated calibration [1], [2]. Programmable interferometric networks and optical switches similarly rely on initialization and compensation for thermal and process drift [3], [4]. Together, these functions form a distributed electronic control plane surrounding the high-speed optical data path.

Although these operations evolve on thermal-tuning or calibration time scales, they are replicated across a large number of optical channels. Their implementation therefore places strong emphasis on control-channel density, reduced off-chip I/O, and

compatibility with the available voltage window. Zero-change MOSFETs can place monitor multiplexing, sample-and-hold, bias-routing, and slow-feedback functions close to the photonic devices without adding dedicated transistor process modules. Previous circuit-level demonstrations of photodiode readout, thermal-actuator control, and programmable-mesh configuration illustrate these application roles [5], [7].

In this context, the measured back-gate response provides a device-level means of aligning the complementary operating window. The shared substrate supplies die-level common-mode threshold positioning, while the lateral gates retain local control of individual devices. This division can support multiplexed monitor readout, storage and routing of phase-setting signals, and slow feedback for temperature and fabrication drift without disturbing the high-speed optical path. The measured polarity-dependent tuning map therefore provides a practical basis for selecting a common substrate bias together with the local gate-voltage range.

IV. Conclusion

Zero-change side-gate MOSFETs fabricated in a 220-nm SOI silicon-photonics process exhibit complementary field-effect operation, but their zero-back-gate thresholds place the native operating window outside the range preferred for compact complementary control. Under a common 100-nA fixed-current definition, the shared substrate shifts the nMOS threshold over 13.30 V and through zero, while the pMOS threshold moves monotonically over 2.96 V across the measured range. The polarity-dependent response identifies the global substrate bias as a common-mode alignment parameter that should be selected together with the local gate operating range. This capability provides a device-level basis for high-impedance monitor selection, phase setting, bias routing, and slow calibration in AI-oriented programmable photonic systems.

Acknowledgment

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